

Linear Algebra Mastery

Vectors

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1 Introduction

A vector is an *ordered* list of finite real number. The list must have both magnitude and direction. Here are some examples of vectors:

1. Location and displacement
2. Colours
3. Images
4. Video
5. Audio
6. Quantity
7. Daily return of stock
8. Stock portfolio and investment
9. Time series and signals
10. Word count and histogram
11. Cash flow in and out an entity or organization

There are different ways to write a vector. In this article, a vector will be a lower case letters with an arrow at the top(e.g. $\vec{v}$), a scalar will be represented by a lower case letter without and arrow at the top(e.g. v) or in special cases with a Greek letter (e.g. β).

2 Vector Operations

2.1 Vector Addition

We add vectors component by component. e.g. $\vec{u} + \vec{v}$

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}$$

Quick check:

With the following vectors $\vec{v} = (1, 3)$ and $\vec{u} = (2, -1)$, compute $\vec{v} + \vec{u}$.

Solution:

$$\vec{v} + \vec{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Properties

1. Commutative: $\vec{v} + \vec{u} = \vec{u} + \vec{v}$
2. Associative: $\vec{a}(\vec{v} + \vec{u}) = (\vec{a} + \vec{v}) + \vec{u}$
3. Adding a zero vector to any vector has no effect e.g. $\vec{v} + 0 = \vec{v}$
4. Subtracting a vector by itself yields a zero vector e.g. $\vec{v} - \vec{v} = 0$

2.2 Scalar Multiplication

A scalar is a *real valued number*, sometimes coefficient of a vector. It's possible to scale a vector in ways we want, but before we get to that, below is a generalized way to scale a vector e.g. $\beta\vec{v}$

Let $\beta = c_1, c_2, \dots, c_n \in \mathbb{R}$ and $\vec{v} \in \mathbb{R}^n$

$$\beta(\vec{v}) = \beta \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \beta v_1 \\ \beta v_2 \\ \vdots \\ \beta v_n \end{pmatrix}$$

Scaling a vector can lead to either of the following scenarios:

1. Stretching vector, this is true if $\beta > 1$
2. Shrinking vector, if $0 < \beta < 1$
3. Flipped vector, if $\beta = -1$
4. Flipped and stretched vector, if $\beta < -1$
5. Collapsed to zero vector, if $\beta = 0$

Quick check:

Suppose $\vec{v} = (3, 2, 1)$, compute $2\vec{v}$.

Solution:

$$2 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}$$

Properties

1. Commutative: $\beta\vec{v} = \vec{v}\beta$, for some scalars β and any vector $\vec{v}$
2. Associative: $(\beta\alpha)\vec{v} = \beta(\alpha\vec{v})$, for some scalars β, α and any vector $\vec{v}$
3. Distributive: $(\beta + \alpha)\vec{v} = \beta\vec{v} + \alpha\vec{v}$

3 Linear Combination

Linear combination scales vectors and add them e.g. $c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n$ give that, $c_i \in \mathbb{R}$ and $\vec{v}_i \in \mathbb{R}^n$.

Quick check:

Given $\vec{v}_1 = (1, 2)$ and $\vec{v}_2 = (3, -1)$, compute $2\vec{v}_1 + 3\vec{v}_2$

Solution:

$$2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3 \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 9 \\ -3 \end{pmatrix} = \begin{pmatrix} 11 \\ 1 \end{pmatrix}$$

Special Combinations

1. $c_1 = c_2 = c_3 = \dots = c_n = 1$, sum of all vectors
2. $c_1 = c_2 = c_3 = \dots = c_n = \frac{1}{n}$, Average of vectors.

4 Span

Span is a set of all linear combinations of a set of vectors. e.g.

$$\text{span}\{\vec{v}_1, \vec{v}_2\} = \{c_1\vec{v}_1, c_2\vec{v}_2 \mid c_i \in \mathbb{R}\}$$

A span of two *parallel* vectors in $\mathbb{R}^2$ is just a line, no new plane is formed, but if the vectors are not parallel, they will span the entire $\mathbb{R}^2$ plane.

Quick check:

Given $\vec{v}_1 = (1, 0)$ and $\vec{v}_2 = (0, 1)$, what is $\text{span}\{\vec{v}_1, \vec{v}_2\}$.

Solution:

$$\text{span}\{\vec{v}_1, \vec{v}_2\} = \text{span}\left\{c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\} \text{ Given } c_1, c_2 \in \mathbb{R}$$

There are two special linear combinations of vectors yielding the sum of a set of vectors and average of the same.

1. Sum of vectors, $\beta_1\vec{v}_1 = \beta_2\vec{v}_2 = \dots = \beta_n\vec{v}_n = 1$, for some vectors $\vec{v}_i \in \mathbb{R}^n$ and any scalars $\beta_i \in \mathbb{R}$.
2. Average of vectors, $\beta_1\vec{v}_1 = \beta_2\vec{v}_2 = \dots = \beta_n\vec{v}_n = \frac{1}{n}$, for $\vec{v}_i \in \mathbb{R}^n$ and $\beta_i \in \mathbb{R}$

5 Linear Independence

A collection or a list of n -vectors is said to be linearly independent if $\beta_1 \vec{v}_1 + \beta_2 \vec{v}_2 + \cdots + \beta_n \vec{v}_n = 0$, for all $\beta_i = 0$.

Quick check:

Given $\vec{v}_1 = (1, 2)$ and $\vec{v}_2 = (3, 5)$, determine if they are linearly independent.

Solution:

Recall: $\beta_1 \vec{v}_1 + \beta_2 \vec{v}_2 + \cdots + \beta_n \vec{v}_n = 0$

$$\beta_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \beta_2 \begin{pmatrix} 3 \\ 5 \end{pmatrix} = 0$$

$$\beta_1 + 3\beta_2 = 0 \tag{1}$$

$$2\beta_1 + 5\beta_2 = 0 \tag{2}$$

$$2\beta_1 + 6\beta_2 = 0$$

$$2\beta_1 + 5\beta_2 = 0$$

$$\implies \beta_2 = 0 \text{ and } \beta_1 = 0$$

6 Linear dependence

A collection of n -vectors is linearly dependent if $\beta_1 \vec{v}_1 + \beta_2 \vec{v}_2 + \cdots + \beta_n \vec{v}_n = 0$, for some scalars β_i not all zeros.

A collection of n -vectors containing the zero vector is always linearly dependent.

Proof:

Let $S = \{\vec{v}_1, \vec{v}_2, \cdots \vec{v}_n\}$ and $\vec{v}_k \in S$ such that $v_k = 0$

Linear combination is given as

$$\beta_1 \vec{v}_1 + \cdots + \beta_k \vec{v}_k + \cdots + \beta_n \vec{v}_n = 0$$

Dividing by β_k

$$\frac{\beta_1}{\beta_k} \vec{v}_1 + \cdots + \vec{v}_k + \cdots + \frac{\beta_n}{\beta_k} \vec{v}_n = 0$$

Finding $\vec{v}_k$

$$\vec{v}_k = - \left(\frac{\beta_1}{\beta_k} \right) \vec{v}_1 + - \left(\frac{\beta_2}{\beta_k} \right) \vec{v}_2 + \cdots + - \left(\frac{\beta_{k-1}}{\beta_k} \right) \vec{v}_{k-1} + - \left(\frac{\beta_{k+1}}{\beta_k} \right) \vec{v}_{k+1} + \cdots + - \left(\frac{\beta_n}{\beta_k} \right) \vec{v}_n$$

From the equation above, it's evident that not all β_i are all equal to zero, hence we can construct a new zero vector, which is redundant.

7 Basis

A basis is a collection of n -vectors that are linearly independent and spans the entire space. Every vector in the space is reachable. Examples of a standard basis:

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Proof:

Suppose we have a collection of $n + 1$ n -vectors, given as $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n, \vec{u}\}$. The set S is linearly dependent, meaning that not all $\beta_i \neq 0$.

If we have

$$\beta_1 \vec{v}_1 + \beta_2 \vec{v}_2 + \dots + \beta_n \vec{v}_n + \beta_{n+1} \vec{v}_{n+1} = 0$$

Then, $\beta_{n+1} = 0$, a contradiction for linear dependence.

8 Machine Learning Application

1. In machine learning, vectors can be represented by NumPy arrays.
2. Stretching, collapsing, or flipping a vector is a common practice while dealing with images, training neural network models.
3. Linear regression models, return their predictions as a linear combination of model weights and the target vector.
4. Multicollinearity causes a model prediction to be unstable. A collection of vectors that are linearly dependent causes multicollinearity.
5. Span tells us the true dimensionality of a set of vectors. A target vector should lie within the span, otherwise no exact solution will exist.